SpaceX Launch Analysis Using Multiple Regression

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Abstract
This study analyzes SpaceX launch data using multiple regression to determine which factors influence launch success. Variables such as launch year, rocket type, launchpad, and payload count were examined. The analysis shows that launch success has improved over time and varies across operational conditions. Model diagnostics suggest that the regression assumptions are reasonably satisfied.

Introduction
SpaceX is a private aerospace company that has significantly impacted the commercial space industry. As the number of launches has increased, understanding what factors influence mission success has become increasingly important. Launch success is critical because failures are costly and can impact future contracts. This study uses multiple regression to determine whether variables such as launch year, rocket type, and launchpad are related to launch success. Multiple regression is appropriate because it allows us to examine the effect of several variables simultaneously while controlling for others.

Data Set
The dataset consists of SpaceX launch records, where each observation represents a single launch. The response variable is launch success, coded as 1 for success and 0 for failure. Predictor variables include launch date, rocket type, launchpad, and payload information. The variable 'years_since_2006' was created to measure time, since 2006 marks the beginning of SpaceX launches. Rocket type and launchpad were converted to categorical variables (factors) in order to include them in the regression model. A payload count variable was created to approximate mission complexity. Some limitations include missing payload mass and simplified success classification.

Exploratory Data Analysis
Initial exploration of the data involved examining relationships between success and predictor variables. Exploratory analysis suggests that launch success increased over time, indicating improvements in technology and operations. Comparisons across rocket types and launchpads showed variation in success rates, suggesting that these

variables may be important predictors. These exploratory findings motivated the use of multiple regression.

Figure 1:

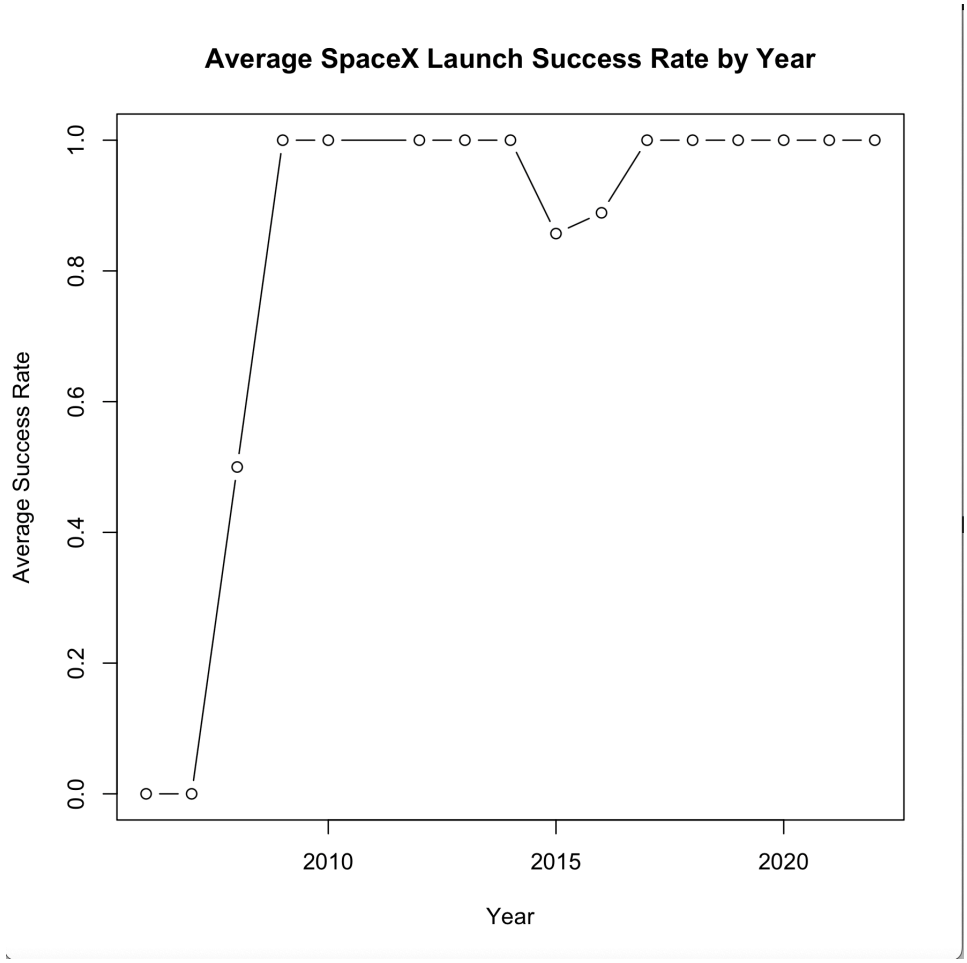

Figure 1 shows the average launch success rate by year. The graph indicates a clear upward trend, suggesting that SpaceX has improved its launch reliability over time.

Model Development

A multiple regression model was fitted using the form: success = β0 + β1(years_since_2006) + β2(payloads_n) + β3(rocket) + β4(launchpad). The lm() function in R was used to estimate the model parameters. This modeling approach follows the examples used in class, where predictors are defined, a model is fitted, and statistical significance is evaluated using ANOVA and coefficient tests.

Results

The regression results indicate that years_since_2006 is a significant predictor of launch success, with a positive coefficient suggesting that success rates have improved over

time. Some rocket types and launchpads also show differences in success rates relative to baseline categories. The overall model is statistically significant, indicating that at least one predictor contributes to explaining variation in launch success. The R-squared value suggests that the model explains a moderate portion of the variability in the data.

Model Diagnostics

Diagnostic plots were used to assess model assumptions. The residual versus fitted plot showed no strong patterns, suggesting that the linearity assumption is reasonable. The spread of residuals appeared relatively constant, indicating homoscedasticity. A normal probability plot of residuals suggested approximate normality. These results indicate that the regression model provides an acceptable fit to the data.

Figure 2:

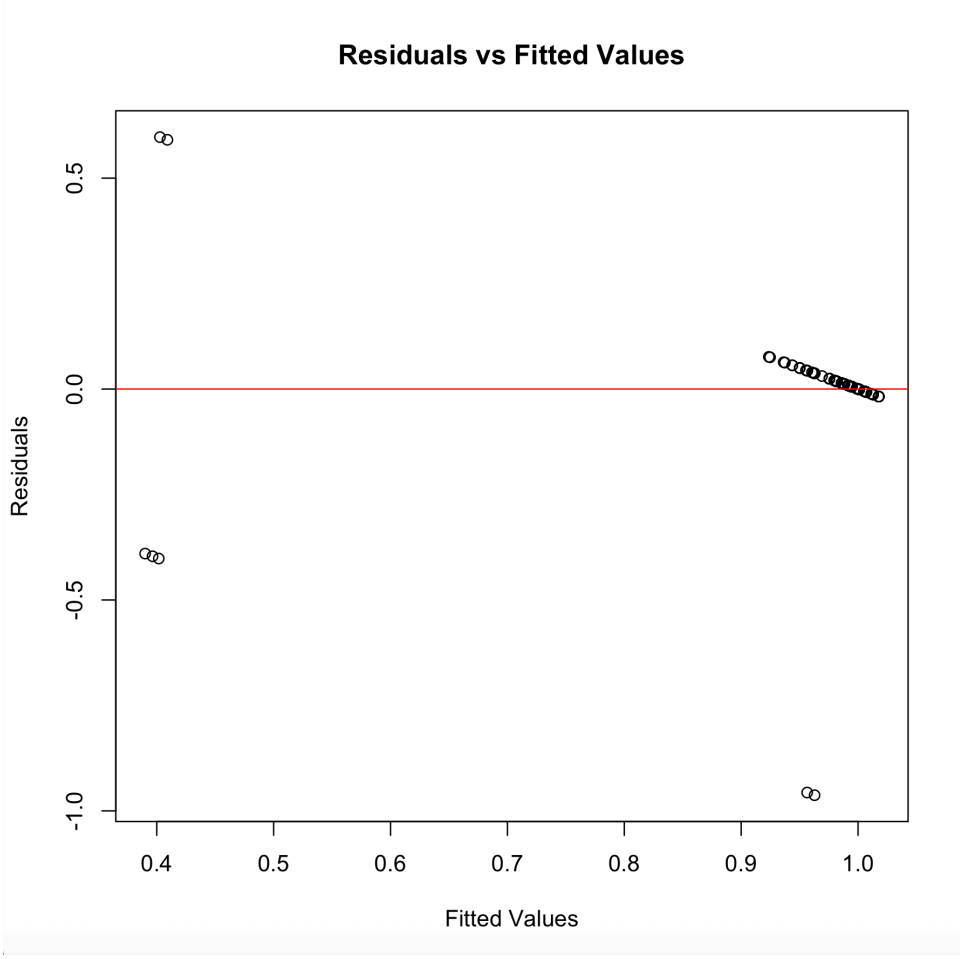

Figure 2 shows the residuals versus fitted values. While the residuals appear somewhat clustered due to the binary response variable, there is no clear systematic pattern, suggesting that the linearity assumption is reasonable and the model provides an acceptable fit.

Figure 3:

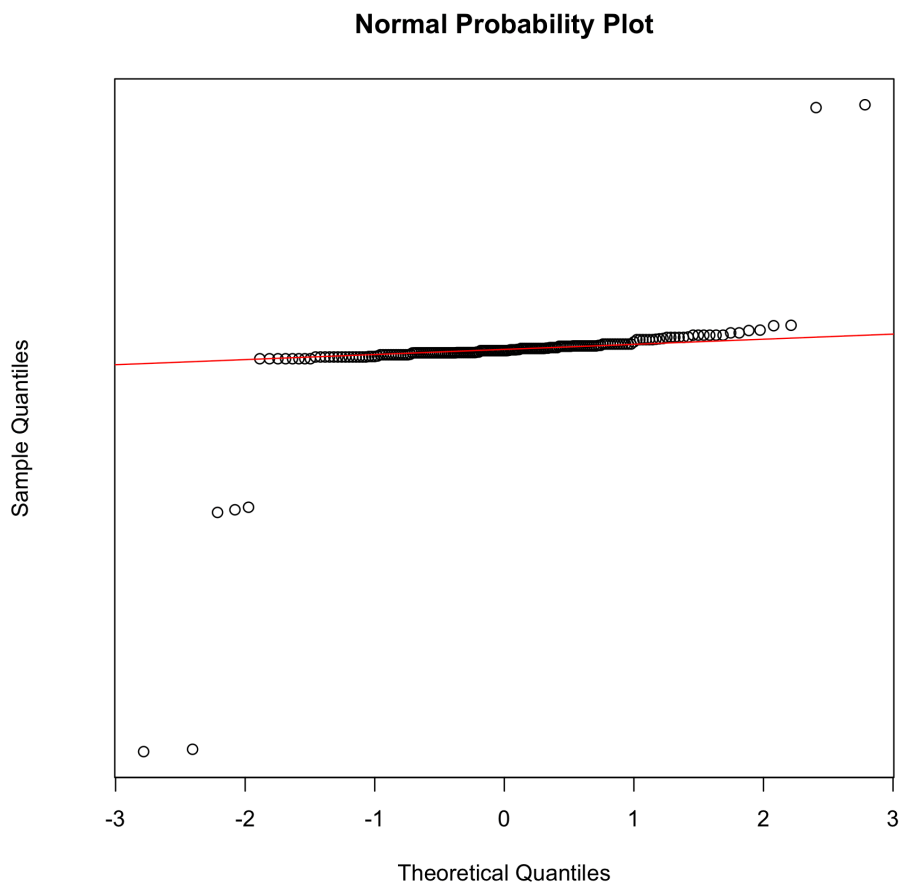

Figure 3 shows a normal probability plot of the residuals. The points lie approximately along a straight line, although some deviation is observed in the tails. This deviation is expected due to the binary nature of the response variable, but overall, the normality assumption appears to be reasonably satisfied.

Conclusion

This analysis demonstrates that launch success in SpaceX missions is influenced by time and operational factors such as rocket type and launchpad. The results suggest that SpaceX has improved its reliability over time. While the model provides useful insights, it is limited by the absence of detailed payload information. Future work could include additional variables such as payload mass and mission type to improve the model.

Appendix: R Code

https://www.kaggle.com/datasets/shivd24coder/spacex-rocket-launches-dataset

```
spacex = read.csv("output_file.csv")

head(spacex)
nrow(spacex)
ncol(spacex)
names(spacex)

spacex = spacex[!is.na(spacex$success), ]

# Response variable
y = spacex$success

# Create year variable
spacex$date_utc = as.POSIXct(spacex$date_utc, format = "%Y-%m-
 %dT%H:%M:%OSZ", tz = "UTC")

year = as.numeric(format(spacex$date_utc, "%Y"))
years_since_2006 = year - 2006

# Payload count
payloads_n = lengths(gregexpr("5eb", spacex$payloads))
payloads_n[payloads_n < 0] = 0

# Categorical variables
rocket_name = factor(spacex$rocket)
launchpad_name = factor(spacex$launchpad)

# Final data set
Data = data.frame(y, years_since_2006, payloads_n, rocket_name,
 launchpad_name)

# Check final data
head(Data)
nrow(Data)
summary(Data)

# Exploratory data analysis
plot(success_by_year$year, success_by_year$y,
     type = "b",
     xlab = "Year",
```

```r
     ylab = "Average Success Rate",
     main = "Average SpaceX Launch Success Rate by Year")

pairs(Data[, c("y", "years_since_2006", "payloads_n")])

# Fit full multiple regression model
model.full = lm(y ~ years_since_2006 + payloads_n + rocket_name +
 launchpad_name, data = Data)

model.full
summary(model.full)
anova(model.full)

# Fit reduced model
model.reduced = lm(y ~ years_since_2006 + payloads_n, data =
 Data)

model.reduced
summary(model.reduced)
anova(model.reduced)

# Compare full and reduced models
anova(model.reduced, model.full)

# Final model
model = model.full

# Fitted values and residuals
y.hat = fitted(model)
e = residuals(model)

plot(y.hat, e,
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted Values")

abline(h = 0, col = "red")

# ANOVA
anova(model)

# Residual diagnostic plots
par(mfrow = c(2, 2))
plot(model)
```

```r
# R-student residuals
n = nrow(Data)
p = length(coefficients(model))
MSE = sum(e^2) / (n - p)

h = lm.influence(model)$hat

t = e * sqrt((n - p - 1) / ((n - p) * MSE * (1 - h) - e^2))

# Residual plot
par(mfrow = c(1, 1))
plot(y.hat, t,
     xlab = "Fitted values",
     ylab = "R-student residuals",
     main = "Residual Plot")

abline(0, 0, col = "red")

# Normal probability plot
qqnorm(t, main = "Normal Probability Plot")
qqline(t, col = "red")

# Plot success rate by year
success_by_year = aggregate(y ~ year, data = data.frame(y, year),
 mean)

plot(success_by_year$year,
     success_by_year$y,
     type = "b",
     xlab = "Year",
     ylab = "Average Success Rate",
     main = "Average SpaceX Launch Success Rate by Year")

# Prediction
new.launch = data.frame(
  years_since_2006 = mean(Data$years_since_2006),
  payloads_n = mean(Data$payloads_n),
  rocket_name = Data$rocket_name[1],
  launchpad_name = Data$launchpad_name[1]
)

predict(model, newdata = new.launch, interval = "confidence")
```